1 Cryptocurrency Risk Regimes: A Comparative Analysis of Volatility and Correlation (2022–2025)

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1.1 Scenario

This analysis quantifies the risk characteristics of digital assets relative to traditional financial benchmarks over a three-year period characterized by significant macroeconomic shifts and crypto-specific market events. The primary objective is to test the "digital gold" narrative and assess the institutional maturity of Bitcoin (BTC) and Ethereum (ETH) by measuring their realized volatility and correlation with the S&P 500 (SPY) and Gold (GLD).

Understanding these metrics is critical for institutional portfolio construction. If crypto assets exhibit dampening volatility and uncorrelated returns during market stress, they serve as effective diversifiers. Conversely, high correlation during drawdowns would indicate they function more as high-beta risk assets.

1.2 Experimentation

The analysis utilizes daily closing price data from December 2022 to December 2025. Data was sourced via the Yahoo Finance API, adjusted for splits and dividends. The core metrics calculated include:

- **Rolling Annualized Volatility:** Calculated using a 30-day sliding window of log returns, scaled by $\sqrt{252}$ to annualize the standard deviation. This captures short-term risk regimes.
- **Rolling Correlation:** A 30-day sliding window correlation coefficient between Bitcoin and the S&P 500 to measure the time-varying interdependence of asset classes.
- **Beta Sensitivity:** The covariance of Bitcoin returns with the S&P 500 divided by the variance of the S&P 500, providing a measure of systematic risk.

Key market events, such as the Silicon Valley Bank (SVB) Crisis and the BTC ETF Approval, were annotated to contextualize structural breaks in the data.

1.3 Analysis and Results

The volatility analysis (Figure 1) highlights a stark contrast in risk profiles between asset classes. Over the observed period, the average annualized volatility

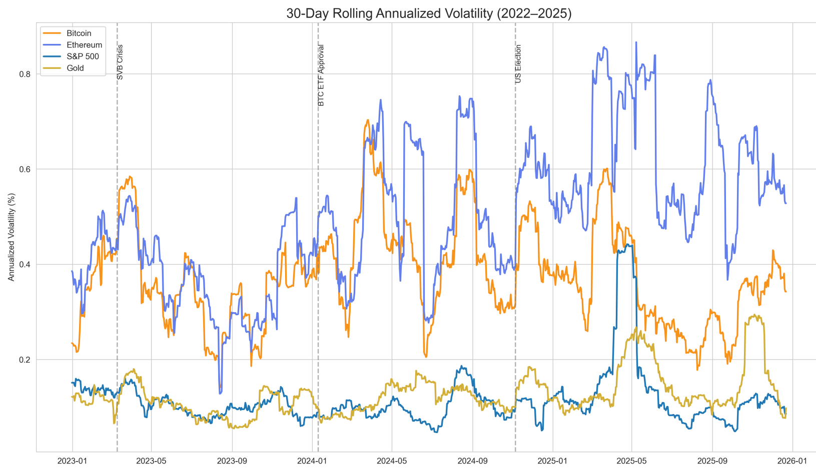

Figure 1: 30-Day Rolling Annualized Volatility Comparison (2022–2025). Note the distinct volatility clustering in digital assets (BTC/ETH) compared to the relatively stable baseline of SPY and Gold.

for Bitcoin was **37.4%** and Ethereum **50.4%**, significantly higher than the S&P 500 (**11.4%**) and Gold (**12.7%**). This results in a volatility ratio of approximately **3.3x** for Bitcoin relative to the equity market.

Notably, volatility regimes in crypto are event-driven. The collapse of SVB in early 2023 triggered a spike in BTC volatility that decoupled from traditional equities, reinforcing the asset's sensitivity to banking sector instability. However, the post-ETF approval period in 2024 shows a gradual compression in volatility spreads, suggesting that institutional inflows may be dampening the erratic price action historically associated with the asset class.

The correlation analysis (Figure 2) reveals that Bitcoin does not maintain a stable relationship with the S&P 500. The correlation coefficient oscillates significantly, ranging from near 0.70 during periods of macro-economic uncertainty (e.g., the U.S. Election cycle) to negative values during idiosyncratic crypto events.

The calculated Beta of **0.89** suggests that, on average, Bitcoin has moved with a sensitivity slightly less than unity relative to the S&P 500 over this specific window. However, the rolling correlation chart demonstrates that this average is misleading; the relationship is highly regime-dependent. During the "risk-on" rallies of late 2023 and 2024, correlation tightened, indicating that Bitcoin was treated by the market as a high-growth technology proxy rather than a non-correlated store of value.

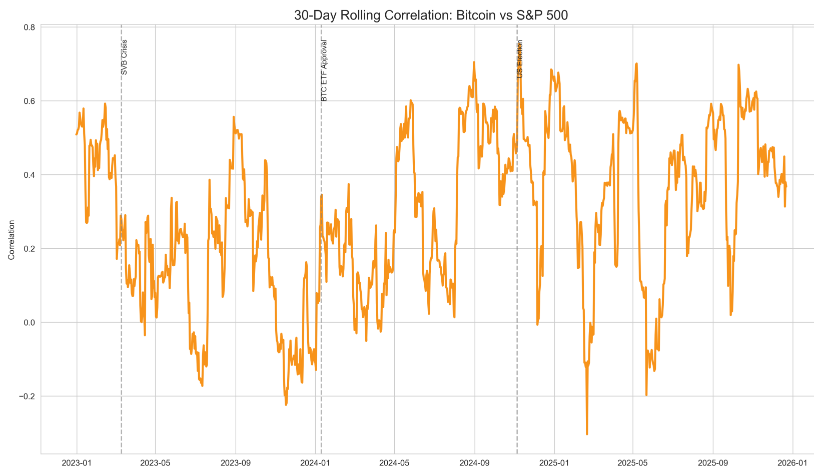

Figure 2: 30-Day Rolling Correlation between Bitcoin and S&P 500. The fluctuating correlation indicates that Bitcoin acts as a non-stationary diversifier.

1.4 Conclusion

This study confirms that while Bitcoin and Ethereum continue to exhibit significantly higher volatility than traditional assets, their market structure is evolving. The decoupling of volatility during banking crises suggests nascent "safe haven" properties, yet the high correlation during equity rallies points to continued classification as a risk asset. For institutional investors, the 3.3x volatility ratio necessitates strict position sizing, but the non-stationary correlation offers opportunistic windows for effective portfolio diversification.